

Online Appendix

Institutional Profiles Under a Common Prudential Framework: Evidence from Brazilian Credit Cooperatives and Banks

Arthur Gomes Nery, Thiago de Oliveira Victorino and Rodrigo Lima Rangel

Organization of Brazilian Cooperatives (OCB System), Brazil

All tables and figures are the authors' calculations from BACEN IF.data, on the analysis sample of the main text unless a note says otherwise. Outcomes are defined in Table 1 of the main text and the specifications in its Section 3.4. The material follows the order of the results: the sample and the main estimates (Tables A1 to A3, Figure A1), the dispersion of the outcomes and the reporting-level split (Tables A4 and A5, Figures A2 and A3), the comparison groups (Tables A6 and A7), the network split (Table A8), the sensitivity checks (Tables A9 to A11) and the de-cumulation of the income fields (Figure A4). The quantile regression coefficients are Table 3 of the main text.

Classification rule

Each outcome in Table 2 of the main text carries a tier. An outcome is *null* unless it is significant at five percent, with one sign, in all four specifications. A significant outcome is *stable* if it stays significant under every winsorisation rule (1/99, 2.5/97.5, 5/95 and none), varies by less than a factor of two across those rules, and has an average quantile coefficient between the twentieth and eightieth percentiles of at least half the conditional mean. Otherwise it is *stable in sign*: the direction holds, but the tails or the shape of the distribution drive the magnitude.

Small-cluster simulation

With six treated clusters drawn at random from the cooperatives and relabelled banks as the null, over 200 draws, the clustered test rejects a true null 16.5 percent of the time at a nominal 5 percent level and 10 percent at a nominal 1 percent level; the restricted wild cluster bootstrap rejects 10 percent and 0.5 percent. Bootstrap rejections at one percent are therefore conservative, while those at five percent still over-reject. The main comparison rests on about 130 treated clusters, where clustered and bootstrap inference agree; the system splits in Table A8 report their treated cluster counts.

Provisioning coverage

The provisioning ratio is observed for 97 percent of cooperative-quarters and 84 percent of bank-quarters in the analysis sample. The gaps concentrate among the low-credit institutions that the comparison-group ladder removes and in 2023, when coverage falls to 75 percent for cooperatives and 66 percent for banks. Re-estimation on the rows where provisioning is observed leaves every surviving sign unchanged (Section 4.6 of the main text).

Table A1: Descriptive statistics

Outcome	Cooperatives		Banks		Observations	
	Mean	Median	Mean	Median	Coop.	Banks
Basel capital ratio (pp)	22.77	20.46	30.96	18.41	3,048	4,170
Leverage	0.8337	0.8493	0.7777	0.8644	3,048	4,184
Return on assets, after tax	0.0288	0.0284	0.0108	0.0116	3,048	4,181
Return on assets, before tax	0.0320	0.0315	0.0202	0.0173	3,048	4,181
Cost-to-income ratio	0.5536	0.5516	1.1561	0.6455	3,048	3,853
Credit portfolio / assets	0.5673	0.5899	0.4020	0.3956	3,048	4,184
Provisioning ratio	0.0546	0.0503	0.0527	0.0322	2,945	3,520
Intermediation margin	0.0839	0.0809	0.0842	0.0516	3,048	4,181
Funding ratio	0.6156	0.6114	0.5694	0.6752	3,048	4,184

Unconditional, analysis sample after winsorisation. Observations are institution-quarters with the outcome observed.

Table A2: Cooperative differential with confidence intervals

Outcome	(1) Raw	(2) Controls	(3) CEM	(4) Trim
Basel capital ratio (pp)	-8.19*** [-13.03, -3.35]	-15.77*** [-21.93, -9.62]	-17.50*** [-26.14, -8.85]	-14.21*** [-22.07, -6.34]
Leverage	0.0561** [0.0211, 0.0910]	0.1220*** [0.0820, 0.1620]	0.1046*** [0.0659, 0.1433]	0.0813*** [0.0415, 0.1211]
Return on assets, after tax	0.0181*** [0.0132, 0.0229]	0.0215*** [0.0143, 0.0286]	0.0179*** [0.0117, 0.0241]	0.0159*** [0.0090, 0.0228]
Return on assets, before tax	0.0118*** [0.0054, 0.0183]	0.0132** [0.0036, 0.0228]	0.0088 [-0.0000, 0.0176]	0.0062 [-0.0036, 0.0160]
Cost-to-income ratio	-0.6025*** [-0.7876, -0.4174]	-0.7923*** [-1.0467, -0.5378]	-0.8385*** [-1.1975, -0.4795]	-0.7522*** [-1.0444, -0.4600]
Credit portfolio / assets	0.1654*** [0.1121, 0.2187]	0.1881*** [0.1286, 0.2477]	0.1701*** [0.0942, 0.2459]	0.1780*** [0.1071, 0.2488]
Provisioning ratio	0.0020 [-0.0087, 0.0126]	-0.0012 [-0.0151, 0.0127]	0.0010 [-0.0135, 0.0156]	0.0009 [-0.0144, 0.0161]
Intermediation margin	-0.0003 [-0.0190, 0.0184]	-0.0190 [-0.0450, 0.0070]	-0.0107 [-0.0334, 0.0121]	-0.0171 [-0.0455, 0.0112]
Funding ratio	0.0461 [-0.0019, 0.0942]	0.0887** [0.0344, 0.1429]	0.0536 [-0.0080, 0.1151]	0.0330 [-0.0281, 0.0940]
Return on assets volatility	-0.0110*** [-0.0150, -0.0069]	-0.0189*** [-0.0229, -0.0148]	-0.0175*** [-0.0221, -0.0130]	-0.0170*** [-0.0220, -0.0120]
Z-score	4.20*** [1.98, 6.42]	6.32*** [4.19, 8.44]	6.14*** [3.57, 8.71]	6.20*** [3.45, 8.95]

Each coefficient with its 95 percent interval beneath it, clustered by institution; HC3 for the two institution-level rows. Stars as in Table 2 of the main text.

Table A3: Conditional mean and median

Outcome	Conditional mean	Conditional median	Ratio	Skewness	Same sign
Basel capital ratio (pp)	-15.77***	-2.18	0.14	4.06	yes
Leverage	0.1220***	0.0283	0.23	-2.70	yes
Return on assets, after tax	0.0215***	0.0171	0.80	-1.68	yes
Return on assets, before tax	0.0132**	0.0144	1.09	-0.13	yes
Cost-to-income ratio	-0.7923***	-0.1796	0.23	4.46	yes
Credit portfolio / assets	0.1881***	0.2448	1.30	-0.34	yes
Provisioning ratio	-0.0012	0.0236	-19.49	3.36	no
Intermediation margin	-0.0190	0.0203	-1.07	3.86	no
Funding ratio	0.0887**	-0.0247	-0.28	-1.07	no
Return on assets volatility	-0.0189***	-0.0133	0.71	1.97	yes
Z-score	6.32***	7.58	1.20	0.85	yes

Controlled specification, cooperatives against banks. Ratio: conditional median over conditional mean.

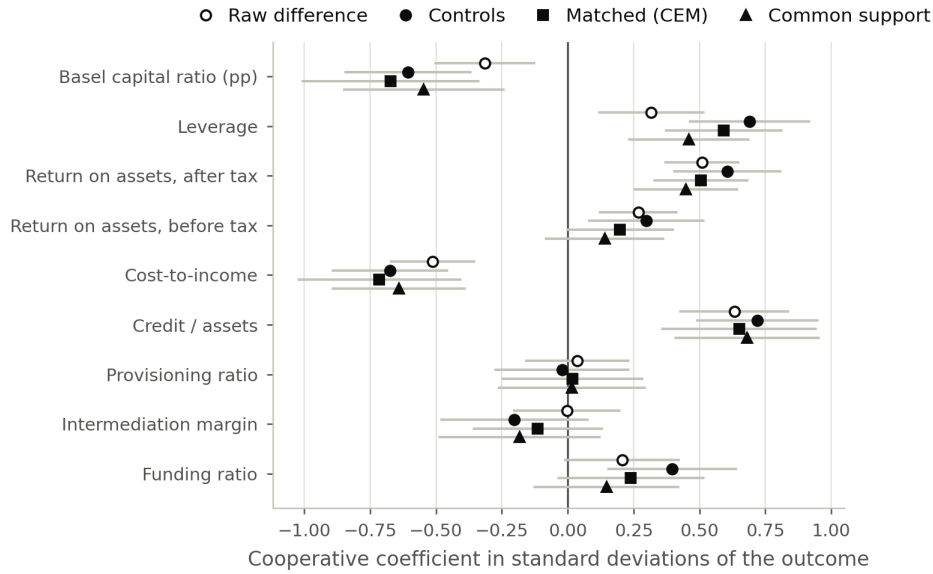


Figure A1: Cooperative coefficient by specification

In standard deviations of each outcome. Intervals are 95 percent, clustered by institution.

Table A4: Dispersion at equal size

Outcome	Residual s.d.		Ratio	Residual IQR	
	Coop.	Banks		Coop.	Banks
Basel Capital Ratio (%)	9.2379	29.7383	0.31	8.3718	18.3313
Leverage (liabilities/assets)	0.0759	0.1788	0.42	0.0793	0.1429
Return on Assets	0.0180	0.0420	0.43	0.0174	0.0225
Return on Assets, pre-tax	0.0185	0.0553	0.33	0.0179	0.0340
Cost-to-Income Ratio	0.1491	1.4751	0.10	0.1859	0.7334
Credit Portfolio / Assets	0.1208	0.3088	0.39	0.1486	0.5459
Provisioning (provisions/gross credit)	0.0243	0.0701	0.35	0.0213	0.0507
Z-score	7.4423	9.3498	0.80	8.5542	11.2177

Residuals from a regression of each outcome on log assets and quarter fixed effects. Ratio: cooperative over bank residual standard deviation; below one, cooperatives are the tighter group. The z-score is institution-level, residualised on mean log assets. Over the seven outcomes jointly, the generalised variance ratio is 0.08.

Table A5: Banks by reporting level

<i>Panel A. Reporting level and the unconditional Basel ratio, analysis sample</i>						
Group	Institutions	Quarters	Mean	Median	p_{90}	Share > 50%
Banks, conglomerate consolidations	87	2,604	22.7	16.8	39.2	6.4
Banks, individual	57	1,580	44.5	24.5	109.1	26.4
Cooperatives (all individual)	130	3,050	22.8	20.5	32.9	1.0

<i>Panel B. Cooperative differential by reporting-level rung, primary winsorisation</i>				
Outcome	(1) Raw	(2) Controls	(3) CEM	(4) Trim
<i>Banks reporting individually</i>				
Basel capital ratio (pp)	-21.77*** (5.26)	-21.42*** (4.81)	-20.87*** (6.41)	-20.65*** (6.54)
Leverage	0.127*** (0.033)	0.120*** (0.025)	0.090*** (0.025)	0.072** (0.027)
Cooperative / bank clusters	128 / 57	128 / 57	128 / 57	119 / 48
<i>Banks reporting as prudential conglomerates</i>				
Basel capital ratio (pp)	0.07 (1.75)	-5.39* (2.72)	-10.35** (4.61)	-5.11 (3.93)
Leverage	0.013 (0.018)	0.082*** (0.024)	0.115*** (0.030)	0.054** (0.023)
Cooperative / bank clusters	128 / 87	128 / 87	128 / 49	104 / 44

<i>Panel C. Basel ratio, cooperative coefficient across the conditional distribution</i>										
Bank rung	q_{10}	q_{20}	q_{30}	q_{40}	q_{50}	q_{60}	q_{70}	q_{80}	q_{90}	OLS
Individual	2.1	0.7	-0.7	-3.0	-5.8	-10.5	-17.1	-25.8	-55.7	-21.4
Conglomerate	3.4	3.1	2.6	2.3	1.7	0.3	-2.7	-7.1	-17.4	-5.4

<i>Panel D. Residual dispersion at equal size, banks reporting individually</i>					
Outcome	Residual s.d.			Residual IQR	
	Coop.	Banks	Ratio	Coop.	Banks
Basel Capital Ratio (%)	10.4318	41.2907	0.25	10.3302	30.0517
Leverage (liabilities/assets)	0.0902	0.2060	0.44	0.0973	0.1987
Return on Assets	0.0183	0.0501	0.37	0.0186	0.0320
Return on Assets, pre-tax	0.0188	0.0606	0.31	0.0194	0.0454
Cost-to-Income Ratio	0.1713	1.3451	0.13	0.2149	0.6632
Credit Portfolio / Assets	0.1171	0.3175	0.37	0.1514	0.6062
Provisioning (provisions/gross credit)	0.0240	0.0914	0.26	0.0201	0.0496
Z-score	7.3375	9.0318	0.81	8.5027	10.4707

Reporting level is the document-type field of IF.data's prudential report (C: conglomerate consolidation; I: individual), taken as the institution's modal value; 25 banks switch level inside the sample. Panel A: winsorised analysis sample. Panel B: the four specifications with the bank side restricted to one level (57 individual, 87 conglomerate), every outcome clipped at the primary sample's winsorisation bounds; clustered standard errors in parentheses, stars from the restricted wild cluster bootstrap (1,999 replications; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$). Panel C: quantile regressions of the Basel ratio, controlled specification. Panel D: residual dispersion as in the dispersion table.

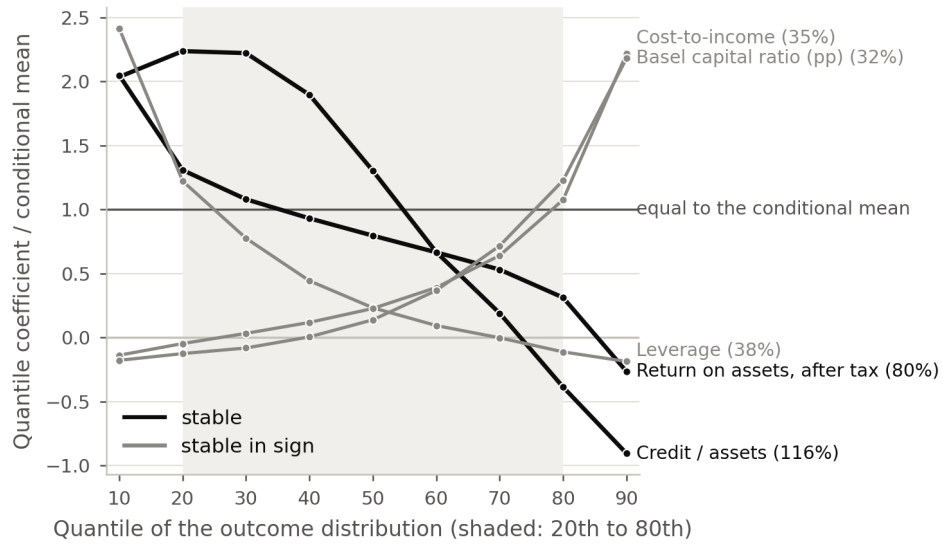


Figure A2: Quantile coefficients relative to the conditional mean
Shaded: 20th to 80th percentile. The percentage beside each label is the average quantile coefficient over that range as a share of the conditional mean.

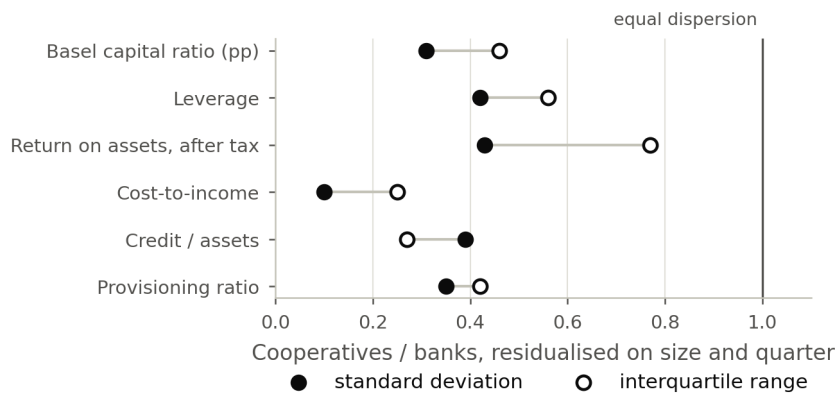


Figure A3: Dispersion ratios at equal size
Cooperative over bank value, residualised on log assets and quarter.

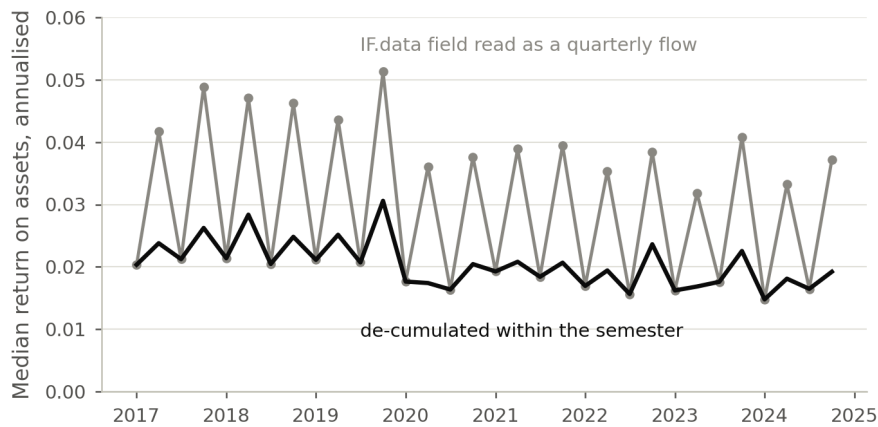


Figure A4: Return on assets before and after de-cumulation
Median across the analysis sample, annualised.

Table A6: Cooperative differential by comparison group

Outcome	All non-cooperatives	Banks	Banks, within region	Commercial	Structural
Basel capital ratio (pp)	-8.27*** (1.52)	-15.77*** (3.14)	-7.02* (3.03)	-17.11*** (4.57)	-7.54** (2.91)
Leverage	0.1280*** (0.0127)	0.1220*** (0.0204)	0.1052*** (0.0276)	0.0881*** (0.0231)	0.0356* (0.0181)
Return on assets, after tax	0.0171*** (0.0032)	0.0215*** (0.0036)	0.0131* (0.0061)	0.0282*** (0.0044)	0.0283*** (0.0039)
Return on assets, before tax	-0.0017 (0.0045)	0.0132** (0.0049)	0.0023 (0.0089)	0.0242*** (0.0055)	0.0267*** (0.0047)
Cost-to-income ratio	-1.8277*** (0.1840)	-0.7923*** (0.1298)	-0.6966* (0.3145)	-0.9740*** (0.1716)	-1.0180*** (0.1991)
Credit portfolio / assets	0.2030*** (0.0200)	0.1881*** (0.0304)	-0.0254 (0.0634)	0.2614*** (0.0287)	0.2425*** (0.0313)
Provisioning ratio	-0.0330*** (0.0059)	-0.0012 (0.0071)	0.0010 (0.0087)	0.0062 (0.0064)	0.0080 (0.0061)
Intermediation margin	-0.0910*** (0.0121)	-0.0190 (0.0133)	-0.0201 (0.0144)	0.0046 (0.0118)	0.0207** (0.0072)
Funding ratio	0.1768*** (0.0199)	0.0887** (0.0277)	0.0548 (0.0369)	0.0299 (0.0315)	-0.0348 (0.0284)
Return on assets volatility	-0.0624*** (0.0044)	-0.0189*** (0.0021)	-0.0170*** (0.0037)	-0.0153*** (0.0022)	-0.0111*** (0.0021)
Z-score	7.79*** (0.87)	6.32*** (1.08)	5.05 (3.19)	7.34*** (1.16)	7.69*** (1.24)
Comparison institutions	483	144	144	105	94

Controlled specification; the within-region column also exact-matches on macro-region. Clustered standard errors in parentheses; HC3 for the two institution-level rows. Rungs as defined in Section 3.2 of the main text.

Table A7: Provisioning by capital tercile

Tercile	Range	Coefficient (clustered SE)	Bootstrap p	Quarters coop. / banks	Mean provisioning	
					Coop.	Banks
<i>Panel A. Terciles of the Basel ratio residualised on log assets and quarter fixed effects</i>						
Bottom	[-40.8, -9.7]	-0.0105 (0.0095)	0.290	1,375 / 796	0.053	0.063
Middle	[-9.7, -0.6]	0.0007 (0.0104)	0.936	1,181 / 1,105	0.054	0.052
Top	[-0.6, 165.1]	0.0109 (0.0139)	0.446	389 / 1,605	0.063	0.047
<i>Panel B. Terciles of the unconditional Basel ratio</i>						
Bottom	[9.8, 16.8]	-0.0081 (0.0086)	0.355	657 / 1,598	0.048	0.058
Middle	[16.8, 23.9]	0.0178*** (0.0056)	0.001	1,344 / 881	0.054	0.039
Top	[23.9, 189.4]	-0.0007 (0.0127)	0.951	944 / 1,027	0.060	0.057

Cooperative coefficient, controlled specification, within each tercile of the pooled sample: Panel A cut on the Basel ratio residualised on log assets and quarter fixed effects, Panel B on the ratio itself. Clustered standard errors in parentheses; p -values and stars from the restricted wild cluster bootstrap (1,999 replications; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$). Quarters with a provisioning ratio.

Table A8: Cooperative differential by system

System	Inst.	Treated	Basel (pp)	Leverage	ROA	Provisioning
Sicredi	106	106	-17.35***	0.1341***	0.0219***	-0.0008
Sicoob	17	17	-10.98*	0.0847*	0.0164*	-0.0007
Independent	6	5	6.90	-0.0313	0.0287*	-0.0165
Cresol	1	–	fewer than five institutions, not estimated			

Controlled specification against the bank group. Stars from the restricted wild cluster bootstrap: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Treated: cooperative clusters with a usable observation; with five, the smallest attainable bootstrap p-value is about 0.03. Sicoob's capital coefficient has a bootstrap p-value of 0.019. The independent return-on-assets coefficient has an asymptotic p-value below 0.001 and a bootstrap p-value of 0.033.

Table A9: Excluding S1 and S2 banks

Outcome	All segments	Excluding S1 and S2
Basel capital ratio (pp)	-15.77***	-16.02***
Leverage	0.1220***	0.1230***
Return on assets, after tax	0.0215***	0.0216***
Return on assets, before tax	0.0132**	0.0133**
Cost-to-income ratio	-0.7923***	-0.7988***
Credit portfolio / assets	0.1881***	0.1873***
Provisioning ratio	-0.0012	-0.0025
Intermediation margin	-0.0190	-0.0197
Funding ratio	0.0887**	0.0877**

Controlled specification. Institutions are excluded on their modal segment under Res. CMN 4.553/2017; S1 and S2 contain no cooperatives. The restricted column compares the 130 cooperatives with 133 banks over 6,882 institution-quarters.

Table A10: Before and after the 2022 capital rules

Outcome	Full sample	2017Q1–2021Q4	2022Q1–2024Q4
Basel capital ratio (pp)	-15.77***	-16.90***	-13.96***
Leverage	0.1220***	0.1337***	0.1033***
Return on assets, after tax	0.0215***	0.0242***	0.0173***
Return on assets, before tax	0.0132**	0.0145*	0.0111*
Cost-to-income ratio	-0.7923***	-0.8533***	-0.6927***
Credit portfolio / assets	0.1881***	0.2046***	0.1629***
Provisioning ratio	-0.0012	-0.0082	0.0104
Intermediation margin	-0.0190	-0.0251	-0.0093
Funding ratio	0.0887**	0.1128***	0.0508
Observations	7,234	4,436	2,798
Cooperatives	130	112	120
Banks	144	141	134

Controlled specification, cooperatives against banks, on the full sample and on each subperiod; the boundary is 3 January 2022, when Resolutions CMN 4.955 and 4.958 took effect. Standard errors clustered by institution. Bootstrap p-values (1,999 replications) agree with the clustered stars in every cell.

Table A11: Stability outcomes before and after taxes

Outcome	Measure	Coefficient	95% interval	Median, coop.	Median, banks
Return volatility	after tax	-0.0462***	[-0.0620, -0.0305]	0.0107	0.0165
	before tax	-0.0268***	[-0.0320, -0.0216]	0.0113	0.0220
Z-score	after tax	7.58***	[5.43, 9.73]	17.73	11.09
	before tax	8.79***	[6.95, 10.63]	17.19	8.63

Cooperative coefficient on one row per institution, controlling for mean log assets, HC3 standard errors; after tax uses line (j), before tax line (g), both de-cumulated and annualised. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.